

Our Clinic — Medical Appointment Coordination

Slide 1 / 5

Team 06 · Problem & Vision

The problem

- Patients rely on **phone/IVR** — long holds, unclear availability, dead ends when a specialty is full
- **Multi-site clinics** (Brooklyn, Midtown, Queens) — hard to find the nearest open slot
- Front desk juggles walk-ins, phone bookings, and reminders without one live view
- Missed appointments when confirmations and reminders are manual

Our vision

One coordinated platform for patients, providers, front desk, and admin — aligned with real healthcare appointment-call data (Kaggle 2024).

Maria

Patient — mobile chat, booking, reminders

Dr. Chen

Provider — calendar & availability

Elena

Front desk — check-in, walk-ins

David

Admin — locations, utilization reports

Requirements

Slide 2 / 5

Functional & non-functional (MVP P0)

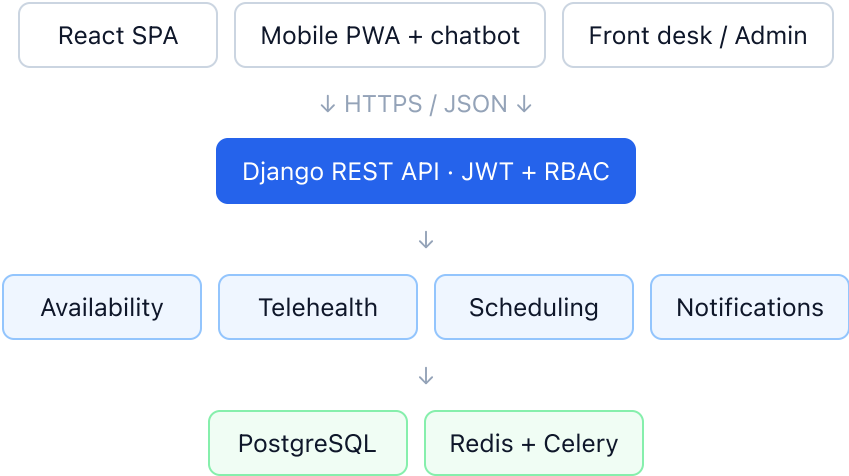
Functional requirements

Epic	User stories	What it must do
Patient booking	US-4.6, US-4.7	Search slots by service; find closest practitioner with availability
Mobile channel	US-4.8	Chatbot — book, nearest doctor, view/cancel appointments
Telehealth fallback	US-4.9	No in-person slots → call doctor or book online video visit
Booking	US-4.2, US-4.3	Book, cancel, reschedule; no double-booking
Front desk	US-5.1–5.3	Walk-in booking, check-in, day schedule across providers
Notifications	US-7.1, US-7.2	Confirmation email + T–24 h reminder (telehealth includes join link)

Non-functional requirements

NFR	Target	NFR	Target
Search latency	p95 ≤ 500 ms	Security	JWT + RBAC
Booking write	p95 ≤ 800 ms	Data source	Kaggle call transcripts
Concurrency	One slot → one appointment	Notifications	Async; booking not blocked by email queue

Layered architecture



Key design decisions

- Empty in-person search → telehealth (not a dead end)
- Same booking pipeline for in-person and video visits
- Async notifications — booking succeeds if email queue is down

Core domain model

Entity	Responsibility
Service	Cardio, General Doctor, Skin Care (patient-friendly labels)
Slot	Bookable unit — modality: in-person or telehealth
Appointment	Status + booking_channel (web, phone, mobile_chat, telehealth)
Provider	Specialty, location, phone, telehealth enabled
NotificationLog	Confirmation & reminder audit trail
CallRecord	Phone/IVR transcripts (Kaggle alignment)

Solution

Slide 4 / 5

Phase 0 prototype — what we built

Deliverables

Component	Deliverable
Mockup	17 screens — patient, provider, desk, admin
Mobile app	Chatbot, Home, Visits tabs
Search	By service + closest practitioner (haversine)
Telehealth	Call doctor + online meeting with join URL
Reminders	T-24 h scheduling + notification log
Mock API	prototype/server.py — availability, notifications

Tech stack

Now (Phase 0)	Planned (Phase 1+)
HTML/CSS/JS + Python mock server	Django 5 + DRF + PostgreSQL + Redis/Celery + React

Demo flow (~60 seconds)

- 1 **Maria · Mobile app** → Chat → Book General Doctor → confirm
- 2 Book **Skin Care** → no in-person slots → Call Dr. Wong or Book online meeting
- 3 **Elena · Front desk** → Check-in → Send reminder
- 4 **Appointments** → Simulate 24 h reminders → see email log

Summary & Next Steps

Value delivered & roadmap

Our Clinic replaces phone tag with self-service booking, smart telehealth when slots are full, and automated reminders — one system for the whole practice.

Benefits

Benefit	How
Fewer calls	Mobile chatbot + online search
No dead ends	Call doctor or video visit when in-person is full
Better utilization	Closest-practitioner search across 3 sites
Fewer no-shows	Confirmation + T-24 h reminders
Evidence-based	Requirements from Kaggle call transcripts

Roadmap

Phase	Scope	Status
0	Mockup, chatbot, telehealth, mock API	✓ Done
1	Django models, JWT auth, seed data	Planned
2	Booking API, React patient/provider UI	Planned
3	Front-desk flows, admin config	Planned
4	Celery email + E2E tests	Planned